



PhD Admission — Università di Trieste, Physics

# Decoherence due to graviton scattering

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Theoretical Physics - Curriculum: Quantum mechanics

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An **open system** interacting with a **gravitational environment**  
**What happens to its coherence?**<sup>[1][2]</sup>

## 1. Entangled state

$$|\Psi(0)\rangle \propto (|S_1\rangle + |S_2\rangle) |E_0\rangle$$

$\Downarrow$

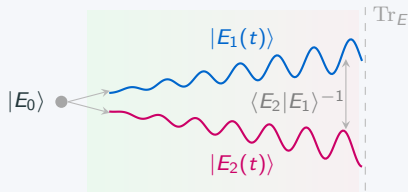
$$|\Psi(t)\rangle \propto |S_1\rangle |E_1(t)\rangle + |S_2\rangle |E_2(t)\rangle$$

## 2. Density matrix

$$\hat{\rho}_S(t) = \text{Tr}_E[|\Psi\rangle \langle\Psi|]$$

$$\hat{\rho}_S(0) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \longrightarrow \hat{\rho}_S(t) = \frac{1}{2} \begin{pmatrix} 1 & \langle E_2|E_1\rangle \\ \langle E_1|E_2\rangle & 1 \end{pmatrix}$$

$$\langle E_2(t)|E_1(t)\rangle \rightarrow 0 \implies \text{decoherence}$$



# Linearized Gravity: Free Theory [3]

## Metric Expansion:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

## Equation of motion

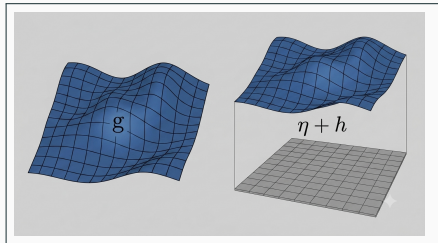
Coordinate invariance:  $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$

$\Rightarrow$  Gauge fixing in vacuum leaves only the propagating degrees of freedom of  $h_{\mu\nu}$

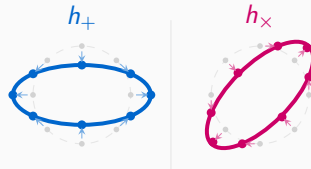
Propagating (TT) solutions:

$$h_{ij}^{\text{TT}}(t, z) = \begin{pmatrix} h_+ & h_\times \\ h_\times & -h_+ \end{pmatrix} \cos[\omega(t - z/c)]$$

TT: Transverse Traceless



## Transverse space-points deformation



## Quantized Linearized Gravity: Interaction with Matter [4][5]

Expand  $\sqrt{-g}\mathcal{L}_M$  around flat space, its structure is *analogous to Scalar QED*. Split  $\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_I$  and quantize:

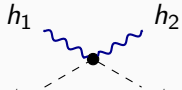
$$\mathcal{H}_I = \underbrace{\frac{1}{2} h_{ij} \partial^i \phi \partial^j \phi}_{\mathcal{O}(h)} + \underbrace{\frac{1}{4} h_{ij} h^{ij} \left( \frac{1}{2} \partial^\mu \phi \partial_\mu \phi - \frac{m^2}{2} \phi^2 \right) + \frac{1}{2} h^{ik} h_k^j \partial_i \phi \partial_j \phi}_{\mathcal{O}(h^2)}$$

The **scattering amplitude** leading order is  $\mathcal{O}(h^2)$ :

$$S_{fi}^{(2)} = \langle \phi(p'), h(k') \mid \hat{S}^{(2)} \mid \phi(p), h(k) \rangle$$

# Matter–Gravity Interactions and Decoherence

Interaction with the  
gravitational environment

$$|S_{fi}^{(2)}|^2 = \left| \begin{array}{c} h_1 \quad h_2 \\ \phi_1 \quad \phi_2 \end{array} \right|^2$$


Decoherence

$$\hat{\rho}_S(0) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \longrightarrow \hat{\rho}_S(t) = \frac{1}{2} \begin{pmatrix} 1 & \langle E_2|E_1 \rangle \\ \langle E_1|E_2 \rangle & 1 \end{pmatrix}$$

- $\rho_{12}(t) \propto \langle E_1(t)|E_2(t) \rangle$

$$S_{fi}^{(2)} \Longrightarrow \rho_{12}(t)$$

# Outlook & Future Perspectives [2][6][7][8]

**Goal:** Provide state of art framework

- Fundamental level analysis
- Practical test of *theories of gravity*

## **Future Perspectives**

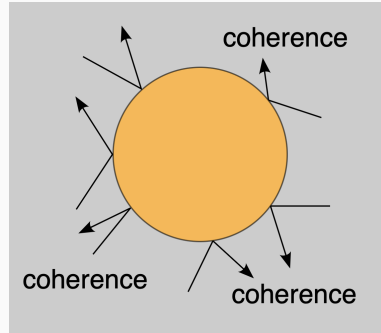
- Collapse models <sup>[7][8]</sup>
- Modern AMO experiments

# References

1. A. Bassi, A. Großardt, H. Ulbricht, *Gravitational Decoherence*, Class. Quantum Grav. **34**, 193002 (2017).
2. E. Calzetta, B. L. Hu, *Nonequilibrium Quantum Field Theory*, Cambridge University Press (2008).
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4. W. Greiner, J. Reinhardt, *Quantum Electrodynamics*, Springer (2009).
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6. A. Gundhi, A. Bassi, *Motion of an Electron through Vacuum Fluctuations*, Phys. Rev. A **108**, 012212 (2023).
7. A. Bassi, K. Lochan, S. Satin, T. P. Singh, H. Ulbricht, *Models of Wave-Function Collapse, Underlying Theories, and Experimental Tests*, Rev. Mod. Phys. **85**, 471 (2013).
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# Thank you for your attention

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## Backup 1 — Graviton–Matter Interaction from $S_M$

Expand  $S_M = \int d^4x \sqrt{-g} \mathcal{L}_M[\phi, g_{\mu\nu}]$  around  $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ .

### Metric expansion to $\mathcal{O}(h^2)$

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + h^\mu_\alpha h^{\alpha\nu}$$
$$\sqrt{-g} = 1 + \frac{h}{2} + \frac{1}{4} \left( \frac{h^2}{2} - h_{\alpha\beta} h^{\alpha\beta} \right)$$

From  $g^{\mu\alpha} g_{\alpha\beta} = \delta^\mu_\beta$  and  
 $\ln \det(-g) = \text{tr} \ln(-g)$ .

### Free term

$$S_M^{(0)} = \int d^4x \left( \frac{1}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{m^2}{2} \phi^2 \right)$$

### Linear vertex — couples $h$ to $T_{\mu\nu}$

$$S_M^{(1)} = -\frac{1}{2} \int d^4x h^{\mu\nu} T_{\mu\nu}$$
$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \eta_{\mu\nu} \left( \frac{1}{2} \partial_\rho \phi \partial^\rho \phi + \frac{m^2}{2} \phi^2 \right)$$

### Seagull vertex — two graviton legs

$$S_M^{(2)} = \int d^4x \left[ \frac{1}{4} \left( \frac{h^2}{2} - h_{\alpha\beta} h^{\alpha\beta} \right) \mathcal{L}_M \right. \\ \left. - \frac{h h^{\mu\nu}}{4} \partial_\mu \phi \partial_\nu \phi + \frac{1}{2} h^{\mu\alpha} h_\alpha^\nu \partial_\mu \phi \partial_\nu \phi \right]$$

In TT gauge ( $h^{0\mu} = 0$ ): time components vanish.

## Backup 2 — Canonical Quantization of the Graviton

### Mode expansion

$$\hat{h}_{\mu\nu}(x) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2\omega_k}} \sum_{\lambda=+, \times} \left( \hat{a}_{\vec{k}, \lambda} e^{-ik \cdot x} \epsilon_{\mu\nu}^{(\lambda)} + \text{h.c.} \right)$$

### Polarization tensors

$$\epsilon_{\mu\nu}^{(+)} = \frac{1}{\sqrt{2}}(u_\mu u_\nu - v_\mu v_\nu)$$

$$\epsilon_{\mu\nu}^{(\times)} = \frac{1}{\sqrt{2}}(u_\mu v_\nu + v_\mu u_\nu)$$

$\{u, v, n\}$  orthonormal;  $n$  = propagation direction.

### TT projector

$$\Lambda_{ij,kl} = \sum_{\lambda} \epsilon_{ij}^{(\lambda)} \epsilon_{kl}^{(\lambda)} = \frac{1}{2}(P_{ik}P_{jl} + P_{il}P_{jk} - P_{ij}P_{kl})$$

### ETCR $\rightarrow$ oscillator algebra

Impose:

$$[\hat{h}_{\mu\nu}(\vec{x}, t), \hat{\pi}_{\alpha\beta}(\vec{x}', t)] = i \delta^{(3)}(\vec{x} - \vec{x}') \delta^{\mu\nu}_{\alpha\beta}$$

Invert via the covariant product

$$\langle A, B \rangle = i \int d^3x A_{\mu\nu}^* \overset{\leftrightarrow}{\partial}_0 B^{\mu\nu}$$

to get  $\hat{a}_{\vec{k}, \lambda} = \langle h(\vec{k}, \lambda), \hat{h} \rangle$ , then compute the commutator:

$$[\hat{a}_{\vec{k}, \lambda}, \hat{a}_{\vec{k}', \lambda'}^\dagger] = \delta_{\lambda\lambda'} \delta^{(3)}(\vec{k} - \vec{k}')$$

## Backup 3 — Seagull Scattering Amplitude

Process:  $\phi(k_3) + h(k_4, \lambda_4) \rightarrow \phi(k_1) + h(k_2, \lambda_2)$ , contact term from  $\hat{S}^{(1)}$  at  $\mathcal{O}(\kappa^2)$ .

### Vertices ( $\mathcal{O}(\kappa^2)$ )

$$V_1 = -\frac{1}{8} h_{\alpha\beta} h^{\alpha\beta} [(\partial^* \phi)^2 - m^2 \phi^2]$$

$$V_2 = \frac{1}{2} h^{\mu\alpha} h_{\alpha}^{\nu} \partial_{\mu} \phi \partial_{\nu} \phi$$

### Mode expansion

Select terms with  $\hat{a}_{k_2, \lambda_2} \hat{a}_{k_4, \lambda_4}$  and  $\hat{a}_{k_1}^{\dagger} \hat{a}_{k_3}$ .

### Momentum conservation

$$\int d^4x e^{i(K_{\text{out}} - K_{\text{in}})x} = (2\pi)^4 \delta^{(4)}(\Delta K)$$

### Tensor structure

$$V_1 : (\epsilon_2 \cdot \epsilon_4)(-k_1 \cdot k_3 + m^2), \quad V_2 : \epsilon_2^{\mu\alpha} \epsilon_{4\alpha}^{\nu} k_{1\nu} k_{3\mu}$$

$$\epsilon(k_2, +) \cdot \epsilon(k_4, +) = \frac{1}{2}(1 + \cos^2 \theta),$$

$$\epsilon(k_2, \times) \cdot \epsilon(k_4, \times) = \cos \theta \quad \theta = \text{scattering angle}$$

$$S_{fi}^{\text{seagull}} = \frac{i \delta^{(4)}(\Delta K)}{(2\pi)^2 \sqrt{16 E_1 E_2 E_3 E_4}} \left[ \frac{1}{2} (\epsilon_2 \cdot \epsilon_4)(-k_1 \cdot k_3 + m^2) + \epsilon_2^{\mu\alpha} \epsilon_{4\alpha}^{\nu} k_{1\nu} k_{3\mu} \right]$$